Large Foundation Models: Mathematics, Algorithms, and Applications

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Learning Objectives

By the end of this course you will be able to:

- Explain what it means to learn a data distribution $p_\theta(\mathbf{x})$ or a conditional $p_\theta(\mathbf{x} \mid \mathbf{c})$.
- Compare the five major families: Autoregressive, VAE, GAN, Normalizing Flows, Diffusion/Flow Matching.
- Relate their training objectives to a unifying divergence/likelihood view.
- Distinguish training vs. sampling costs and typical latency/quality trade-offs.
- Choose an appropriate model family for a task, and see how world models connect to these tools.
- Understanding how the reinforcement learning works.
- Construct your own Models.

Agenda

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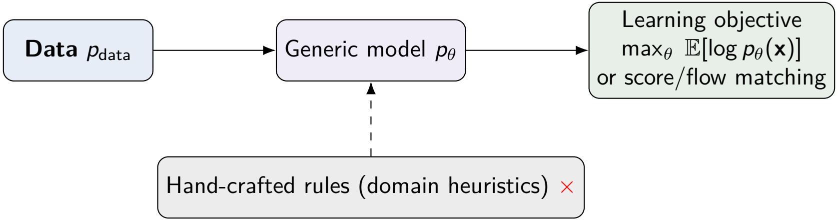
Preliminary: The Bitter Lesson (Sutton)

Thesis. Hand-crafted domain knowledge scales poorly; progress comes from *general-purpose* methods that learn directly from data with lots of compute. Design **mathematical learning algorithms** so models *acquire* the distribution, rather than hard-coding it.

Implications for deep generative modeling

- Prefer scalable objectives: maximize $\log p_\theta(\mathbf{x})$ or minimize a divergence $D(p_{\text{data}} \| p_\theta)$; use variational bounds/score objectives.
- Use generic, data-driven representations (Transformers, CNNs, U-Nets, invertible nets) $+$ self-supervision.
- Scale data/compute; reduce bespoke heuristics and narrow domain rules ($\times$ brittle, hard to scale).



Definition and Why It Matters

Generative models learn a distribution $p(\mathbf{x})$ (or $p(\mathbf{x} \mid \mathbf{c})$) over complex data (images, text, audio, video, 3D, proteins), enabling:

- **Synthesis** (sample new $\mathbf{x}$), **imputation**, **editing**, **control**, **compression**.
- As world-model primitives for **simulation**, **planning**, **data augmentation**.

Two big families of objectives:

1. **Likelihood/variational** training (maximize $\log p_\theta(\mathbf{x})$ or its bound).
2. **Adversarial/implicit** training (match data distribution without tractable likelihood).

Notation & Setup

- Data: $\mathcal{D} = \{\mathbf{x}^{(i)}, \mathbf{c}^{(i)}\}_{i=1}^{N}$, variables may be continuous *or* discrete.
- Parameters: θ; latent variables (when present): $\mathbf{z}$; tokenizer/codebook: $\mathcal{Z}$.
- Objectives: maximize $\log p_\theta$, minimize divergence $D(p_{\text{data}} \| p_\theta)$, or match a transport/flow.
- Inference artifacts: decoders, samplers, inverse maps; report both *train cost* and *sample latency*.

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A Unifying Objective View

Family	Canonical objective (sketch)
Autoregressive	$\max_\theta \sum_i \log p_\theta(x_i \mid x_{<i})$ (exact NLL)
VAE	ELBO: $\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x} \mid \mathbf{z})] - \mathrm{KL}[q_\phi(\mathbf{z} \mid \mathbf{x}) \| p(\mathbf{z})]$
GAN	$\min_G \max_D$ JS/f-divergence matching (implicit likelihood)
Flows	$\log p_\theta(\mathbf{x}) = \log p(\mathbf{z}) + \log\lvert\det \partial f / \partial \mathbf{x}\rvert$
Diffusion/Score	Denoising/score matching; flow/consistency matching

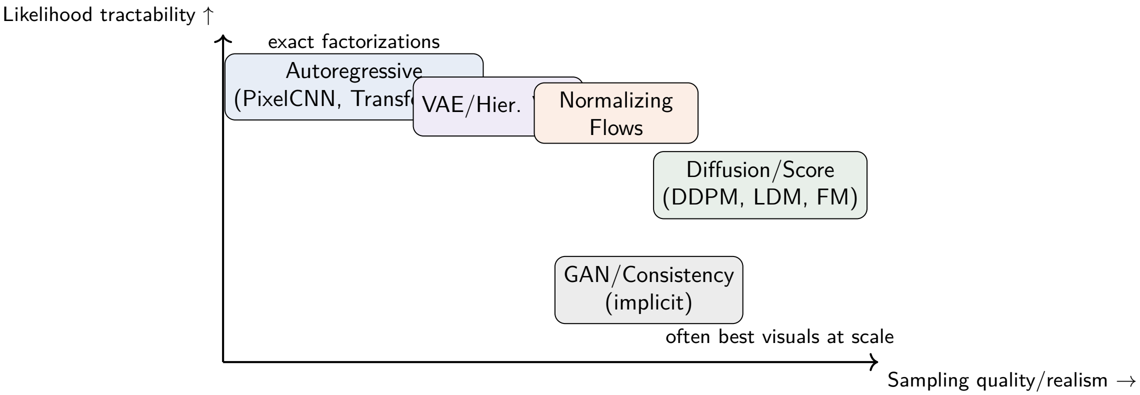
Takeaway: same destination (match p_θ to data), different parameterizations and training signals.

Training vs. Sampling Pipelines

- **Autoregressive:** train once (teacher forcing); sample sequentially (slow, exact likelihood).
- **Diffusion:** train denoiser/score; sample iteratively (few-step via distillation/consistency/flow matching).
- **Flows:** train invertible map; sample in one inverse pass (exact likelihood).
- **VAE:** train encoder/decoder; sample via prior $p(\mathbf{z})$ and decode.
- **GAN:** min-max training; fast single-pass sampling.

Objective → Training → Model Artifact → Sampler/Latency

A Landscape View



Auto regressive modeling

Factorization:

$$p(\mathbf{x}) = \prod_{i=1}^{D} p\big(x_i \mid x_{<i}\big), \quad \text{with an order over dimensions/tokens.}$$

- Image: raster-scan pixels (PixelRNN/CNN); Text/Code: token-by-token (Transformers);
- Audio: next-sample prediction (WaveNet); Vision tokens via VQ-VAE + Transformer.

Training: teacher forcing, **objective** $\sum_i \log p_\theta(x_i \mid x_{<i})$.

Tiny Graphical Intuition

$$x_1 \to x_2 \to x_3 \to x_4$$

Left-to-right factorization
and teacher forcing

- **Pros:** exact likelihood, strong modeling of local dependencies, state-of-the-art in text/code.
- **Cons:** slow sampling (sequential), exposure bias; 2D structure needs careful ordering.

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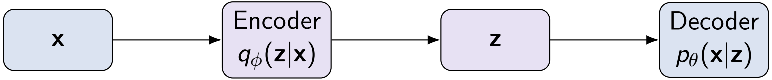
Masked vs. AR, and Speed-ups

- **Masked language modeling (MLM)** learns to fill blanks; not strictly AR but can be combined (e.g., *prefix-decoding*, parallel decoding with iterative refinement).
- **Parallel/Speculative decoding, KV caching, medusa heads** reduce latency.
- **Discretization pipelines:** VQ-VAE/tokenizers + AR over tokens for images/audio/video.

VAE: Idea and Objective



Encoder–Decoder with latent z

$$\log p_\theta(\mathbf{x}) \geq \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\mathrm{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \parallel p(\mathbf{z}))}_{\text{regularization}}$$

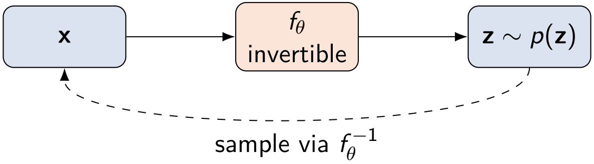
Notes: hierarchical VAEs, β-*VAE*, vector-quantized VAEs (VQ-VAE), diffusion/AR decoders.

GANs: Adversarial Training

$$\min_G \max_D \; \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]$$

- **Pros:** often sharp, realistic samples; once trained, fast sampling.
- **Cons:** no tractable likelihood; training instabilities; mode collapse; harder conditioning.

Normalizing Flows: Invertible Transformations

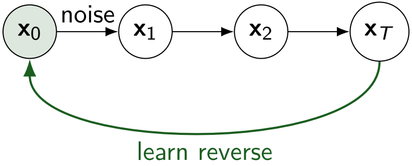


Log-likelihood

$$\log p_\theta(\mathbf{x}) = \log p(\mathbf{z}) + \log \left| \det \frac{\partial f_\theta(\mathbf{x})}{\partial \mathbf{x}} \right|, \quad \mathbf{z} = f_\theta(\mathbf{x}).$$

Notes: RealNVP/Glow (coupling), spline flows, CNFs/Neural ODEs.

Diffusion models: Forward and Reverse Processes



Parameterizations & losses

- **Noise (ε)-prediction**:

$$\mathcal{L}_{\varepsilon} = \mathbb{E}_{\mathbf{x}_0,t,\varepsilon}\left[w_t \left\|\varepsilon - \varepsilon_\theta(\mathbf{x}_t, t)\right\|^2\right], \quad \mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\varepsilon.$$

- x_0-**prediction**:

$$\mathcal{L}_{x_0} = \mathbb{E}_{\mathbf{x}_0,t,\varepsilon}\left[w_t \left\|\mathbf{x}_0 - \hat{\mathbf{x}}_{0,\theta}(\mathbf{x}_t, t)\right\|^2\right], \quad \varepsilon_\theta = \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t}\,\hat{\mathbf{x}}_{0,\theta}}{\sqrt{1-\bar{\alpha}_t}}.$$

- **Score model (DSM)**:

$$\mathcal{L}_{\mathrm{DSM}} = \mathbb{E}_{t,\mathbf{x}_0}\,\mathbb{E}_{\mathbf{x}_t\sim q(\cdot|\mathbf{x}_0)}\left[\lambda(t)\left\|s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}_t}\log q(\mathbf{x}_t \mid \mathbf{x}_0)\right\|^2\right].$$

Flow Matching

Train a velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ moving $p_0 \to p_1$ by the ODE

$$\frac{d\mathbf{x}}{dt} = \mathbf{v}_\theta(\mathbf{x}, t), \quad \mathbf{x}(0) \sim p_0, \ \mathbf{x}(1) \sim p_1 \approx p_{\text{data}}.$$

Loss (conditional flow matching)

$$\mathcal{L}_{\text{CFM}} = \mathbb{E}_{t\sim\mathcal{U}(0,1)}\, \mathbb{E}_{\mathbf{x}_0\sim p_0,\ \mathbf{x}_1\sim p_1}\, \mathbb{E}_{\mathbf{x}_t=(1-t)\mathbf{x}_0+t\mathbf{x}_1}\Big[\, \|\mathbf{v}_\theta(\mathbf{x}_t, t) - (\mathbf{x}_1 - \mathbf{x}_0)\|^2 \,\Big].$$

General path form (with path density π_t and target drift $\mathbf{u}_t$):

$$\mathcal{L} = \mathbb{E}_{t,\mathbf{x}_0,\mathbf{x}_1,\mathbf{x}_t\sim\pi_t(\cdot|\mathbf{x}_0,\mathbf{x}_1)}\Big[\, \|\mathbf{v}_\theta(\mathbf{x}_t, t) - \mathbf{u}_t(\mathbf{x}_0, \mathbf{x}_1, t)\|^2 \,\Big].$$

Perks: simple targets; few-step sampling; latent-friendly.

Converging Lines

- **Tokenizer + AR + Diffusion:** VQ-VAE/LDM token spaces with AR planning/control.
- **Flow Matching Rectified Flow:** train a velocity field $\mathbf{v}_\theta(\mathbf{x}, t)$ to move noise$\to$data with 1st-order ODE sampling.
- **Consistency Models:** distilled one-step or few-step samplers by self-consistency constraints.
- **Mean Flow:** train an averaged velocity for one step generation.
- **Multimodal, Video, 3D/4D:** text-image-audio-video-geometry; world-model style conditional generation.

Takeaway: boundaries blur; choose representations (continuous vs. discrete tokens), objectives, and samplers to fit task/latency.

How We Measure

- **Likelihood / BPD**: exact (AR/flows) or bounds (VAE); diffusion often uses NLL surrogates.
- **Sample Quality**: FID, KID, IS; CLIP-score for text-image alignment.
- **Diversity/Coverage**: precision&recall for distributions; recall of rare modes.
- **Task Utility**: downstream performance, human eval.

Decision Guide: Which Family Should I Choose?

- **Need exact likelihood / lossless compression?** $\Rightarrow$ Autoregressive or Flows.
- **Top visual fidelity / editing?** $\Rightarrow$ Diffusion (latent), consider consistency/rectified/flow matching for speed.
- **Discrete tokens (text/code) or controllable sequences?** $\Rightarrow$ Autoregressive (Transformer) + tokenizers for images/audio.
- **Low-latency sampling at inference?** $\Rightarrow$ GANs/consistency/flow-matched diffusion; Flows when feasible.
- **Structured latent space / downstream control?** $\Rightarrow$ VAE/Hierarchical VAE (+ AR or diffusion decoders).

Practical Trade-offs

Family	Likelihood	Quality	Speed (sample)	Control	Stability
Autoregressive	✓	✓(text/code)	×	✓	✓
VAE	bound	mixed	✓	✓	✓
GAN	×	✓	✓	mixed	mixed
NFlows	✓	mixed	mixed	✓	✓
Diffusion	bound	✓	mixed→✓	✓	✓

Takeaway: Disrete, latent space, mode collapse, inverse design with exact log likelihood, high quality samples with few-steps.

Applications (Examples)

- **Images/Design**: product ideation, content creation, inpainting/outpainting.
- **Text/Code**: LLMs for writing, agents, program synthesis.
- **Audio/Speech/Music**: TTS, music generation, audio restoration.
- **Video/World Models**: controllable video, simulation for planning.
- **3D/Geometry**: NeRF/mesh diffusion, 3D asset pipelines.
- **Science**: molecules/proteins/materials; weather/climate surrogates.

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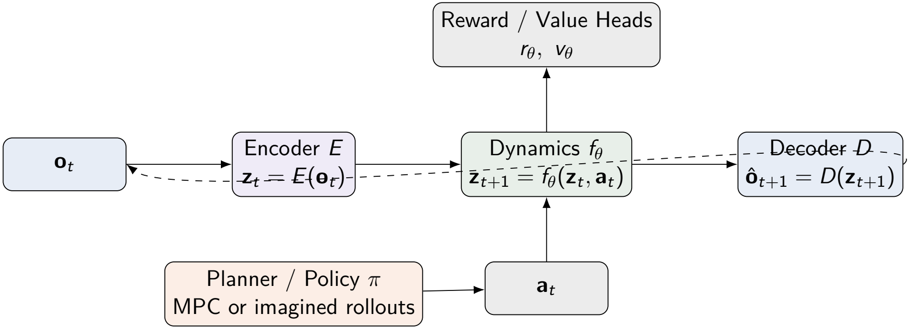
Positioning: From Generative Models to World Models

- **Observation model:** generative $p(\mathbf{o}_t \mid \mathbf{z}_t)$ often shares tech with diffusion/AR/VAEs.
- **Dynamics:** latent transition $f_\theta(\mathbf{z}_{t+1} \mid \mathbf{z}_t, \mathbf{a}_t)$ (Dreamer/JEPA-style) or direct video simulators.
- **Control:** plan over imagined rollouts (MPC) or learn policy $\pi(\mathbf{a}_t \mid \mathbf{z}_t)$.
- **Bridges:** AR for tokens (language/code); diffusion for frames; flows/consistency/flow-matching for speed.

World Models: A Compact Overview (2025)

- **Goal:** learn $p(\mathbf{o}_{t+1} \mid \mathbf{o}_t, \mathbf{a}_t)$ or latent dynamics f_θ; plan via MPC or policy over imagined trajectories.
- **Two flavors:** (i) action-conditioned *video/world simulators*; (ii) representation-first models (e.g., latent JEPA/Dreamer-style) for control/planning.
- **Training signals:** next-frame/latent prediction, contrastive/self-supervision, auxiliary reward/value heads.
- **Eval:** physical plausibility, action-consistency, controllability, and task success; long-horizon memory.
- **Open issues:** causality/counterfactuals, persistent memory/state, sim-to-real, safety.



Takeaway: World Models are systems that learn the dynamics of an environment to plan and control actions by imagining future outcomes, but still face key challenges in causality, memory, and real-world application.

Key Messages

1. **Five pillars:** AR, VAE, GAN, Flows, Diffusion. Each optimizes a different compromise.
2. **Autoregression** remains the gold standard for discrete sequences; for continuous media, tokenization + AR is powerful.
3. **Diffusion/Score-based** dominates visual fidelity; distillation and flow-matching narrow the latency gap.
4. **Hybrids** are the 2025 norm: choose representation + objective + sampler for your latency/quality/control budget.

Pointers (non-exhaustive)

- **AR:** PixelRNN/CNN; GPT-style Transformers for text/code; VQ-VAE + Transformer for images/audio.
- **VAE:** Kingma and Welling (2014); Hierarchical/β-VAE; VQ-VAE.
- **GAN:** Goodfellow et al. (2014); WGAN, StyleGAN series.
- **Flows:** RealNVP, Glow; Neural ODEs; spline flows.
- **Diffusion/Score:** Sohl-Dickstein (2015), DDPM, DDIM, LDM; consistency/rectified/flow-matching; DPM-Solver.

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